42.4: Blood components function in exchange, transport, and defense

As you read in Concept 42.1, the fluid transported by an open circulatory system is continuous with the fluid that surrounds all of the body cells and thus has the same composition. In contrast, the fluid in a closed circulatory system can be much more specialized, as is the case for the blood of vertebrates.

Blood Composition and Function

Vertebrate blood is a connective tissue consisting of cells suspended in a liquid matrix called **plasma**. Separating the components of blood using a centrifuge reveals that cellular elements (cells and cell fragments) occupy about 45% of the volume of blood **(Figure 42.16)**. The remainder is plasma.

Figure 42.16

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The composition of mammalian blood.
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Two tables, one summarizing the constituents of plasma and their major functions, and the other summarizing the cell types, numbers and functions of cellular elements.

Centrifuged blood separates into three layers: plasma, leukocytes and platelets, and erythrocytes.

First table, plasma 55%:

Constituent	Major functions
Water	Solvent
Ions (blood electrolytes) include sodium, potassium, calcium, magnesium, chloride, and bicarbonate	Osmotic balance, pH buffering, and regulation of membrane permeability
Plasma protein: albumin	Osmotic balance, pH buffering
Plasma protein: Immunoglobulins (antibodies)	Defense
Plasma protein: apolipoproteins	Lipid transport
Plasma protein: fibrinogen	Clotting

Substances transported by blood include nutrients (such as glucose, fatty acids, vitamins), waste products of metabolism, respiratory gases (O_2 and CO_2), and hormones	

Second table, cellular elements 45%:

Cell type	Number (per micro liter (cubic millimeters) of blood	Functions
Leukocytes (white blood cells): basophils, lymphocytes, eosinophils, neutrophils, and monocytes	5000 to 10,000	defense and immunity
Platelets	250,000 to 400,000	blood clotting
Erythrocytes (red blood cells)	5,000,000 to 6,000,000	transport of O_2 and some CO_2

Plasma

Dissolved in the plasma are ions and proteins that, together with the blood cells, function in osmotic regulation, transport, and defense. Inorganic salts in the form of dissolved ions are an essential component of the blood. Some buffer the blood, while others help maintain osmotic balance. In addition, the concentration of ions in plasma directly affects the composition of the interstitial fluid, where many of these ions have a vital role in muscle and nerve activity. For plasma to serve all of these functions, the concentrations of plasma electrolytes must remain within specific set points.

Like dissolved ions, plasma proteins such as albumins act as buffers against pH changes and help maintain the osmotic balance between blood and interstitial fluid. A number of plasma proteins have additional functions. Immunoglobulins, or antibodies, combat viruses and other foreign agents that invade the body (see Figure 43.10). Apolipoproteins escort lipids, which are insoluble in water and can travel in blood only when bound to proteins. Fibrinogens act as clotting factors that help plug leaks when blood vessels are injured. (The term *serum* refers to blood plasma from which these clotting factors have been removed.)

Plasma also contains many other substances in transit, including nutrients, metabolic wastes, respiratory gases, and hormones. Plasma has a much higher protein concentration than interstitial fluid, although the two fluids are otherwise similar. (Capillary walls, remember, are not very permeable to proteins.)

Cellular Elements

Blood contains two classes of cells: red blood cells, which transport O_2, and white blood cells, which function in defense (see Figure 42.16). Also suspended in blood plasma are **platelets**, cell fragments that are involved in the clotting process.

Erythrocytes

Red blood cells, or **erythrocytes**, are by far the most numerous blood cells. Their main function is O_2 transport, and their structure is closely related to this function. Human erythrocytes are small disks (7–8 μm in diameter) that are biconcave—thinner in the center than at the edges. This shape increases surface area, enhancing the rate of diffusion of O_2 across the plasma membrane. Mature mammalian erythrocytes lack nuclei. This unusual characteristic leaves more space in these tiny cells for **hemoglobin**, the iron-containing protein that transports O_2 (see Figure 5.18). Erythrocytes also lack mitochondria and generate their ATP exclusively by anaerobic metabolism. Oxygen transport would be less efficient if erythrocytes were aerobic and consumed some of the O_2 they carry.

Despite its small size, an erythrocyte contains about 250 million molecules of hemoglobin (Hb). Because each molecule of hemoglobin binds up to four molecules of O_2, one erythrocyte can transport about 1 billion O_2 molecules. As erythrocytes pass through the capillary beds of lungs, gills, or other respiratory organs, O_2 diffuses into the erythrocytes and binds to hemoglobin. In the systemic capillaries, O_2 dissociates from hemoglobin and diffuses into body cells.

In **sickle-cell disease**, an abnormal form of hemoglobin (HbS) polymerizes into aggregates. Because the concentration of hemoglobin in erythrocytes is so high, these aggregates are large enough to distort the erythrocyte into an elongated, curved shape that resembles a sickle. This abnormality results from an alteration in the amino acid sequence of hemoglobin at a single position (see Figure 5.19).

Sickle-cell disease significantly impairs the function of the circulatory system. Sickled cells often lodge in arterioles and capillaries, preventing delivery of O_2 and nutrients and removal of CO_2 and wastes. Blood vessel blockage and resulting organ swelling can result in severe pain. In addition, sickled cells frequently rupture, reducing the number of red blood cells available for transporting O_2. The average life span of a sickled erythrocyte is only 20days—one-sixth that of a normal erythrocyte. The rate of erythrocyte loss outstrips their production rate. Short-term therapy includes replacement of erythrocytes by blood transfusion; long-term treatments are generally aimed at inhibiting aggregation of HbS.

Leukocytes

The blood contains five major types of white blood cells, or **leukocytes**. Their function is to fight infections. Some are phagocytic, engulfing and digesting microorganisms and debris from the body's own dead cells. Other leukocytes, called lymphocytes, mount immune responses against foreign substances (as we'll discuss in Concepts 43.2 and 43.3). Normally, 1 μL of human blood contains about 5,000–10,000 leukocytes; their

numbers increase temporarily whenever the body is fighting an infection. Unlike erythrocytes, leukocytes are also found outside the circulatory system, patrolling interstitial fluid and the lymphatic system.

Platelets

Platelets are pinched-off cytoplasmic fragments of specialized bone marrow cells. They are about 2–3 μm in diameter and have no nuclei. Platelets serve both structural and signaling functions in blood clotting.

Stem Cells and the Replacement of Cellular Elements

Erythrocytes, leukocytes, and platelets all develop from stem cells that are dedicated to replenishing the body's blood cell populations. As described in Concept 20.3, a **stem cell** can reproduce indefinitely, dividing mitotically to produce one daughter cell that remains a stem cell and another that adopts a specialized function. The stem cells that produce the cellular elements of blood cells are located in the red marrow inside bones, particularly the ribs, vertebrae, sternum, and pelvis. As they divide and self-renew, these stem cells give rise to two sets of progenitor cells with a more limited capacity for self-renewal **(Figure 42.17)**. One set, the lymphoid progenitors, produces lymphocytes. The other set, the myeloid progenitors, produces all other white blood cells, red blood cells, and platelets.

Figure 42.17

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Differentiation of blood cells.
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Lymphoid stem cells (B cells and T cells) and Myeloid stem cells (erythrocytes, platelets, neutrophils, basophils and eosinophils) derive from stem cells in the bone marrow.

Cell divisions of stem cells in bone marrow give rise to two specialized sets of cells. One set—the lymphoid progenitor cells—gives rise to immune cells called lymphocytes, primarily B and T cells. The second set—myeloid progenitor cells—gives rise to other immune cells, red blood cells (erythrocytes), and cell fragments called platelets.

Tree diagram shows the derivation of erythrocytes, leukocytes and platelets from lymphoid cells and myeloid cells after both differentiate from stem cells in bone marrow. They are as follows.

From lymphoid stem cells.

- B cells (lymphocytes).
- T cells (lymphocytes).

From myeloid stem cells.

- Erythrocytes (red blood cells).
- Neutrophils (leukocytes).
- Basophils (leukocytes).
- Monocytes (leukocytes).
- Eosinophils (leukocytes).
- Platelets (cell fragments).

Throughout a person's life, stem cells replace the worn-out cellular elements of blood. Erythrocytes are the shortest-lived, circulating for only 120 days on average before being replaced. A feedback mechanism sensitive to O2 level controls erythrocyte production. If the O2 level falls, the kidneys synthesize and secrete a hormone called **erythropoietin (EPO)** that stimulates the generation of more erythrocytes.

Today, EPO produced by recombinant DNA technology is used to treat disorders such as *anemia*, a condition of lower-than-normal erythrocyte or hemoglobin level that decreases the oxygen-carrying capacity of the blood. Some athletes inject themselves with EPO to increase their erythrocyte level. Because this practice is banned by most major sports organizations, runners, cyclists, and other athletes caught using EPO-related drugs have forfeited their records and been banned from future competitions.

Blood Clotting

When blood vessels are broken by an injury such as a small cut or scrape, a chain of events ensues that quickly seals the break, halting blood loss and exposure to infection. The key mechanical event in this response is coagulation, the conversion of the liquid components of blood into a solid—a blood clot.

In the absence of injury, the coagulant, or sealant, circulates in an inactive form called fibrinogen. Blood clotting begins when injury exposes the proteins in a broken blood vessel wall to blood constituents. The exposed proteins attract platelets, which gather at the site of injury and release clotting factors. These clotting factors trigger a cascade of reactions leading to the formation of an active enzyme, *thrombin*, from an inactive form, prothrombin **(Figure 42.18)**. Thrombin in turn converts fibrinogen to fibrin, which aggregates into threads that form the framework of the clot. Any mutation that blocks a step in the clotting process can cause hemophilia, a life-threatening disease characterized by excessive bleeding and bruising from even minor cuts and bumps (see Concept 15.2).

Figure 42.18

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Blood clotting.
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Platelets adhere to fibers in connective tissue following an injury. Platelet plug prevents blood loss. All but the smallest plugs are reinforced by fibrin clots.

Steps are as follows.

1. The clotting process begins when the endothelium of a vessel is damaged, exposing connective tissue in the vessel wall to blood. Platelets adhere to collagen fibers in the connective tissue and release a substance that makes nearby platelets sticky.
2. The platelets form a plug that provides immediate protection against blood loss.
3. Unless the break is very small, this plug is reinforced by a fibrin clot.

Fibrin clot formation: Clotting factors (result from: platelets, damaged cells, plasma (includes calcium, vitamin K) released from the clumped platelets or damaged cells mix with clotting factors in the plasma, forming an enzymatic cascade that converts a plasma protein called prothrombin to its active form, thrombin. Thrombin itself is an enzyme that catalyzes the final step of the clotting process, the conversion of fibrinogen to fibrin. The threads of fibrin become interwoven into a clot. In an image, red blood cells are caught in threads of fibrin (scale: 5 micrometers).

As shown in Figure 42.18, clotting involves a positive feedback loop. Initially, the clotting reactions convert only some of the prothrombin at the clot site to thrombin. However, thrombin itself stimulates the enzymatic cascade, leading to more conversion of prothrombin to thrombin and thus driving clotting to completion.

Anticlotting factors in the blood normally prevent spontaneous clotting in the absence of injury. Sometimes, however, clots form within a blood vessel, blocking the flow of blood. Such a clot is called a **thrombus** (plural, *thrombi*). We'll explore shortly how thrombi form and the dangers they pose.

Cardiovascular Disease

Each year, cardiovascular diseases—disorders of the heart and blood vessels—kill more than 750,000 people in the United States. These diseases range from minor disturbances of vein or heart valve function to life-threatening disruptions of blood flow to the heart or brain.

Atherosclerosis, Heart Attacks, and Stroke

Healthy arteries have a smooth inner lining that reduces resistance to blood flow. However, damage or infection can roughen the lining and lead to **atherosclerosis**, the hardening of the arteries by accumulation

of fatty deposits. A key player in the development of atherosclerosis is cholesterol, a steroid that is important for maintaining normal membrane fluidity in animal cells (see Figure 7.5).

In atherosclerosis, damage to the arterial lining results in *inflammation*, the body's reaction to injury. Leukocytes are attracted to the inflamed area and begin to take up lipids, including cholesterol. A fatty deposit, called a plaque, grows steadily, incorporating fibrous connective tissue and additional cholesterol. As the plaque grows, the walls of the artery become thick and stiff, and the obstruction of the artery increases. If the plaque ruptures, a thrombus can form in the artery **(Figure 42.19)**, potentially triggering a heart attack or a stroke.

Figure 42.19

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Atherosclerosis.
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Arterial endothelium surrounds the lumen, which is nearly filled with a large thrombus and plaque.

In atherosclerosis, thickening of an arterial wall by plaque formation can restrict blood flow through the artery. If a plaque ruptures, a thrombus can form, further restricting blood flow. Fragments of a ruptured plaque can also travel via the bloodstream and become lodged in other arteries. If the blockage is in an artery that supplies the heart or brain, the result could be a heart attack or stroke, respectively.

A **heart attack**, also called a *myocardial infarction*, is the damage or death of cardiac muscle tissue resulting from blockage of one or more coronary arteries, which supply oxygen-rich blood to the heart muscle. The coronary arteries are small in diameter and therefore especially vulnerable to obstruction by atherosclerotic plaques or thrombi. Such blockage can destroy cardiac muscle quickly because the constantly beating heart muscle requires a steady supply of O2. If a large enough portion of the heart is affected, the heart will stop beating. Such cardiac arrest causes death if a heartbeat is not restored within minutes by cardiopulmonary resuscitation (CPR) or some other emergency procedure.

A **stroke** is the death of nervous tissue in the brain due to a lack of O2. Strokes usually result from rupture or blockage of arteries in the neck or head. The effects of a stroke and the individual's chance of survival depend on the extent and location of the damaged brain tissue. If a stroke results from arterial blockage by a thrombus, rapid administration of a clot-dissolving drug may help limit the damage.

Although atherosclerosis often isn't detected until critical blood flow is disrupted, there can be warning signs. Partial blockage of the coronary arteries may cause occasional chest pain, a condition known as angina pectoris. The pain is most likely to be felt when the heart is laboring under stress, and it signals that part of the heart is not receiving enough O2. An obstructed artery may be treated surgically, either by inserting a

mesh tube called a stent to expand the artery **(Figure 42.20)** or by transplanting a healthy blood vessel from the chest or a limb to bypass the blockage.

Figure 42.20

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Inserting a stent to widen an obstructed artery.
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Placement of a stent widens an obstructed artery and increases blood flow through the artery.

Placement of a stent widens an obstructed artery and increases blood flow through the artery as follows.

1. A stent and a balloon (the stent is around the balloon) are inserted into an obstructed artery (in diagram, plaque is present on arterial walls).
2. Inflating the balloon expands the stent, widening the artery.
3. The balloon is removed, leaving the stent in place. Increased blood flow through the widened artery results.

Risk Factors and Treatment of Cardiovascular Disease

Cholesterol travels in blood plasma mainly in particles that consist of thousands of cholesterol molecules and other lipids bound to a protein. One type of particle—**low-density lipoprotein (LDL)—delivers cholesterol to cells for membrane production. Another type—high-density lipoprotein** ** (HDL)**—scavenges excess cholesterol for return to the liver. Individuals with a high ratio of LDL to HDL are at substantially increased risk for atherosclerosis.

Although the tendency to develop particular cardiovascular diseases is inherited, it is also strongly influenced by lifestyle. For example, exercise decreases the LDL/HDL ratio, reducing the risk of cardiovascular disease. In contrast, consumption of certain processed vegetable oils called *trans fats* and smoking increase the LDL/HDL ratio. For many individuals at high risk, treatment with drugs called statins can lower LDL levels and thereby reduce the risk of heart attacks. In the **Scientific Skills Exercise**, you can interpret the effect of a genetic mutation on blood LDL level.

Scientific Skills Exercise

Making and Interpreting Histograms

AP Science Practices 2, 4, 5, 6

Does Inactivating the PCSK9 Enzyme Lower LDL Levels?Researchers interested in genetic factors affecting susceptibility to cardiovascular disease examined the DNA of 15,000 individuals. They found that 3% of the individuals had a mutation that inactivates one copy of the gene for PCSK9, a liver enzyme. Because mutations that *increase* the activity of PCSK9 are known to *increase* the level of LDL cholesterol in the blood, the researchers hypothesized that *inactivating* mutations in this gene would *lower* the LDL level. In this exercise, you will interpret the results of an experiment they carried out to test this hypothesis.

How the Experiment Was Done Researchers measured the LDL cholesterol level in blood plasma from 85 individuals with one copy of the *PCSK9* gene inactivated (the study group) and from 3,278 individuals with two functional copies of the gene (the control group).

Medical chart.

Data from the Experiment

Two histograms of P C S K 9 gene data.

Two histograms plot the percent of individuals versus milligrams per deciliter of plasma LDL cholesterol for a control group and a study group. The study group consists of individuals with an inactivating mutation in one copy of the P C S K 9 gene. The control group consists of individuals with two functional copies of the P C S K 9 gene. The data is represented by the following table.

Group	Plasma L DL cholesterol in milligrams per deciliter, 0 to 25	26 to 50	51 to 75	76 to 100	101 to 125	126 to 150	151 to 175	176 to 200	201 to 225	226 to 250	251 to 275
Control group	0%	1%	4%	13%	23%	23%	18%	10%	5%	2%	1%
Study group	0%	4%	31%	23%	21%	13%	2%	1%	2%	0%	2%

Data from J. C. Cohen et al., Sequence variations in *PCSK9*, low LDL, and protection against coronary heart disease, *New England Journal of Medicine* 354:1264–1272 (2006).

INTERPRET THE DATA

- The results are presented using a variant of a bar graph called a *histogram*. In a histogram, the variable on the x-axis is grouped into ranges. The height of each bar in this histogram reflects the percentage of samples that fall into the range specified on the x-axis for that bar. For example, in the top histogram, about 4% of individuals studied had plasma LDL cholesterol levels in the 25–50 mg/dL (milligrams per deciliter) range. Add the percentages for the relevant bars to calculate the percentage of individuals in the study and control groups that had an LDL level of 100 mg/dL or less. (For additional information about histograms, see the Scientific Skills Review in Appendix D.)
- Compare the two histograms. Do you find support for the researchers' hypothesis? Explain.
- What if instead of graphing the data, the researchers had compared the range of concentrations for plasma LDL cholesterol (low to high) in the control and study groups? How would their conclusions have differed?
- What does the fact that the two histograms overlap as much as they do indicate about the extent to which PCSK9 determines plasma LDL cholesterol level?
- Comparing these two histograms allowed researchers to draw a conclusion regarding the effect of *PCSK9* mutations on LDL cholesterol levels in blood. Consider two individuals with a plasma LDL level cholesterol of 160 mg/dL, one from the study group and one from the control group.
- What do you predict regarding their relative risk of developing cardiovascular disease?
- Explain how you arrived at your prediction. What role did the histograms play in making your prediction?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

BBC Video

The Impact of Smoking

image

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image

Play

The recognition that inflammation plays a central role in atherosclerosis and thrombus formation is also influencing the treatment of cardiovascular disease. For example, aspirin, which inhibits the inflammatory response, has been found to help prevent the recurrence of heart attacks and stroke.

Hypertension (high blood pressure) is yet another contributor to heart attack and stroke. According to one hypothesis, chronic high blood pressure damages the endothelium that lines the arteries, promoting plaque formation. The usual definition of hypertension in adults is a systolic pressure above 140 mm Hg or a diastolic pressure above 90 mm Hg. Fortunately, hypertension can often be prevented and controlled by quitting any use of tobacco products, changing the diet, increasing exercise, taking medication, or a combination of these approaches.